NCERT Summary: Topic 05: Perfect Competition & Theory of the Firm — Raw Notes

Source Integration

- NCERT Class 12 Introductory Microeconomics, Chapter 4: The Theory of the Firm under Perfect Competition

Core Factual & Conceptual Architecture

1. Structural Characteristics of Perfect Competition

- Atomistic Market Structure: Large number of buyers and sellers where no single agent can alter market price.
- Homogeneous Product: Identical goods across all firms; buyers are indifferent about which firm supplies the good.
- Free Entry and Exit: Absence of legal, technological, or financial barriers to entry or exit.
- Perfect Information: Symmetric knowledge where buyers and sellers have complete awareness of prevailing market prices and technology.
- Behavioral Consequence: Price-taking behavior. The firm cannot sell above the market price and has no reason to sell below it, facing a horizontal demand curve.

2. Revenue Curves of a Price-Taking Firm

- Total Revenue (TR): Calculated as Price multiplied by Quantity sold. Forms a straight line originating from the origin with slope equal to price.
- Average Revenue (AR): Total revenue divided by output, which equals price.
- Marginal Revenue (MR): Extra revenue obtained from selling one more unit.
- Equality Identity: For any price-taking enterprise, Price = Average Revenue = Marginal Revenue.

3. Conditions for Profit Maximisation

- Profit is maximized where the gap between Total Revenue and Total Cost is at its highest:
  - First-Order Condition: Marginal Cost equals Marginal Revenue (MC = MR = Price).
  - Second-Order Condition: Marginal Cost must be rising at that point of intersection.

4. Derivation of the Short-Run Supply Curve

- Market Price at or above Minimum Average Variable Cost (AVC): The firm produces where Short-Run Marginal Cost equals price along the rising section of the Marginal Cost curve.
- Market Price below Minimum Average Variable Cost (AVC): The firm produces zero output because continuing operations would produce variable losses on top of fixed overhead.
- Short-Run Shut-Down Point: The lowest point of the Average Variable Cost curve. Falling below this price halts production immediately.
- Short-Run Supply Locus: The rising portion of the Short-Run Marginal Cost curve at and above the minimum point of Average Variable Cost.

5. Normal Profit, Super-Normal Profit, and Break-Even Point

- Normal Profit: The baseline income required to keep the entrepreneur in business, treated as an implicit cost within total cost.
- Super-Normal Profit: Financial returns realized above normal profit.
- Break-Even Point: The point where market price equals the minimum of Short-Run Average Total Cost. Here, the firm earns normal profit and zero super-normal profit.
- Operational Buffer Zone: When price sits between minimum AVC and minimum SAC, the firm operates at an economic loss but continues producing in the short run to pay off part of its fixed overhead.

6. Long-Run Competitive Supply

- In the long run, all inputs are variable and fixed costs cease to exist.
- Long-Run Supply Curve: The rising section of the Long-Run Marginal Cost curve at and above the minimum point of the Long-Run Average Cost curve.
- Long-Run Exit Point: The minimum point of the Long-Run Average Cost curve. If price falls below this, firms exit the market permanently.

7. Industry Supply & Responsiveness

- Market Supply Curve: Found through the horizontal summation of individual firms' supply curves across all prices.
- Price Elasticity of Supply: Measures how quantity supplied responds to changes in market price, moving positively along an upward-sloping supply line.

✍ Notes & Updates